

NCIT AI Workshop Handbook

Internal Knowledge Document – RAG Demonstration

1. About the Workshop

The RAG (Retrieval-Augmented Generation) Application Development Workshop 2026 is organized by the Computer Application Circle (CAC) to introduce students to modern AI application development. The workshop focuses on building applications that can answer questions based on custom documents instead of relying solely on the knowledge already present in a language model.

Participants will learn the complete workflow of a basic RAG system, including document loading, text splitting, embedding generation, vector database creation, retrieval, and response generation using a Large Language Model (LLM).

2. Workshop Details

The Workshop is happening in 4th July from 11 am at block H room 134.

Target Audience: Undergraduate students interested in Artificial Intelligence, Machine Learning, and Python development.

Mode: Hands-on practical workshop.

3. Learning Objectives

By the end of this workshop, participants will be able to:

- Understand the concept of Retrieval-Augmented Generation (RAG).
 - Build a complete RAG pipeline from scratch.
 - Create embeddings from custom documents.
 - Store and retrieve embeddings using Chroma Vector Database.
 - Connect an LLM with retrieved context to generate accurate answers.
 - Understand common challenges such as hallucination, chunking strategies, and retrieval quality.
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4. Technologies Used

The workshop demonstrates the following technologies:

- Python

- LangChain
 - Chroma Vector Database
 - Hugging Face Embeddings
 - Google Gemini API
 - Streamlit
 - python-dotenv
 - PDF document loaders
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5. RAG Pipeline Overview

The application developed during the workshop follows these steps:

1. Load documents.
 2. Split documents into smaller chunks.
 3. Generate vector embeddings.
 4. Store embeddings inside Chroma.
 5. Accept a user's question.
 6. Retrieve the most relevant document chunks.
 7. Send the retrieved context along with the question to the language model.
 8. Display the generated answer.
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6. Speaker Information

The session is conducted by Aakriti Dahal.

The speaker works in the fields of cloud computing, artificial intelligence, developer communities, and open-source technologies. The session emphasizes practical implementation and encourages participants to build and experiment with AI applications independently.

7. Prerequisites

Participants are expected to have:

- A personal laptop
- Python installed
- Stable internet connection
- A code editor such as Visual Studio Code
- Basic familiarity with command-line tools

No prior experience with LangChain or vector databases is required.

8. Workshop Rules

Participants should:

- Follow along during the hands-on session.
 - Ask questions whenever they encounter difficulties.
 - Avoid sharing API keys publicly.
 - Keep their software updated before the workshop.
 - Respect the learning environment and collaborate with fellow participants.
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9. Computer Application Circle (CAC)

The Computer Application Circle (CAC) is the official student technical club under the BCA Department at NCIT.

The club organizes:

- Technical workshops
- Hackathons
- Knowledge-sharing sessions
- AI and cloud computing events
- Open-source contribution activities
- Project showcases
- Community discussions

Its goal is to provide students with opportunities to gain practical technical experience beyond the classroom.

10. Additional Notes

This handbook serves as the knowledge source for the workshop's RAG demonstration. Participants will upload this document into the application and ask questions related to its contents.

The responses generated by the application depend only on the information contained in this handbook, demonstrating how Retrieval-Augmented Generation can provide accurate and context-aware answers.

11. Frequently Asked Questions

Q: What is Retrieval-Augmented Generation?

A: It is a technique that combines information retrieval with a language model so that answers are generated using external documents.

Q: Which programming language is used?

A: Python is used throughout the workshop.

Q: Which vector database is used?

A: Chroma Vector Database.

Q: Which embedding model is used?

A: Hugging Face Embeddings.

Q: Which language model is demonstrated?

A: Google Gemini.

Q: Is prior AI experience required?

A: No. Basic Python knowledge is sufficient.

Q: Will participants build a complete application?

A: Yes. Participants will build a simple end-to-end RAG application during the workshop.

Q: What type of document is used in the demonstration?

A: A PDF document is used as the knowledge base for retrieval.